Structural model for recruitment of RIT1 to the LZTR1 E3 ligase: evidences from an integrated computational approach

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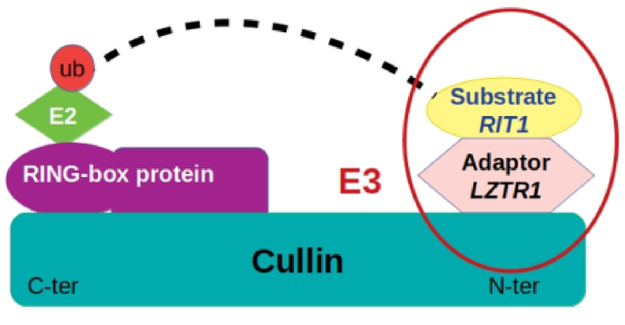
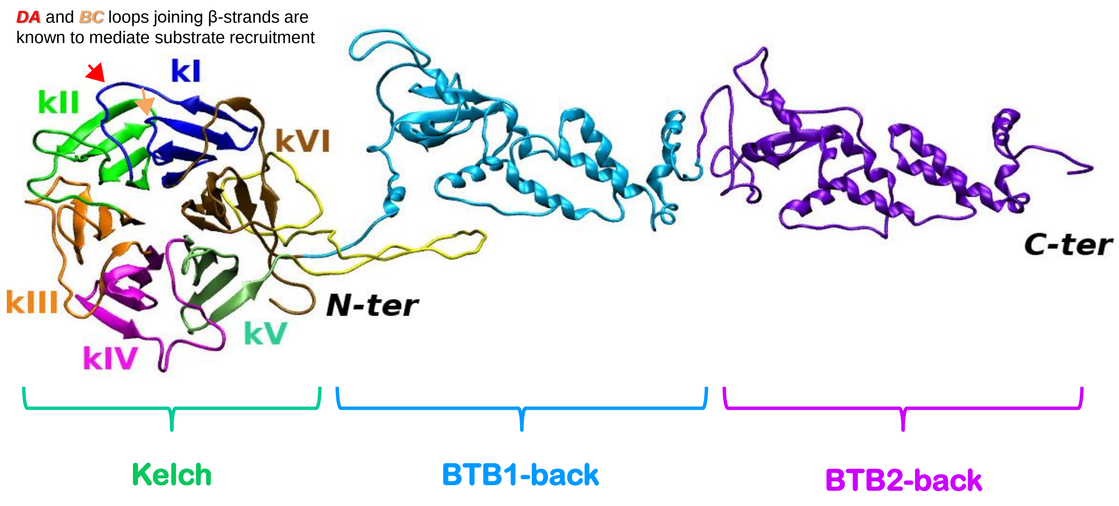
Leucine-zipper transcription Regulator 1 (**LZTR1**) is a highly mutated tumor suppressor gene, involved in the pathogenesis of several cancers and developmental disorders.[1] In proteasomal degradation, it acts as an **adaptor protein** responsible for the recognition and recruitment of substrates to be ubiquitinated in **Cullin3-RING ligase E3** (CRL3) machinery.
A small Ras-related GTPase protein, **RIT1**, has been recently identified by mass spectroscopy to **interact with LZTR1**.[2] Moreover, large efforts mutational analyses pointed to the role of **disease-associated LZTR1 mutations** in RAS/MAPK signaling pathway.[3-5]
No experimental three-dimensional structure is available for the multi-domain LZTR1 protein, a BTB-Kelch family member, with the Kelch propeller domain acting as the substrate recognition region. Here we propose the **interaction model** between adaptor LZTR1 and substrate RIT1 applying an integrated computational approach, build on molecular modelling and docking techniques.[6-7] By MD-based analyses, we also characterize differential **conformational dynamics** of the multi-domain LZTR1, offering interesting implications on the mechanistic role of specific point mutations.[8]



Scheme of Cullin-RING Ubituitin E3 Ligase assembly

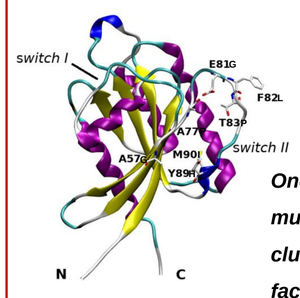
Modelling LZTR1-RIT1 interaction

1. LZTR1 3D structure by homology modelling

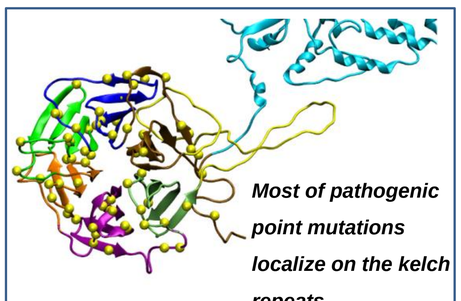


Sequence alignments returns a Kelch six-bladed (aka repeats) β-propeller motif at the N-term and two BTB-back domains at the C-term. A multi-template model of the full-length LZTR1 was built by Modeller, using the Kelch domain of the thiocyanate-forming protein from Thlaspi Arvense (pdb: 5A10) for the Kelch domain and the human SPOP-BTB domain (pdb: 4J8Z) for the BTB domains.
Full-length LZTR1 folds in an elongated structure made by three well-separated structural blocks.

RIT1 is a small GTP-binding protein shown by MS experiments to interact with LZTR1

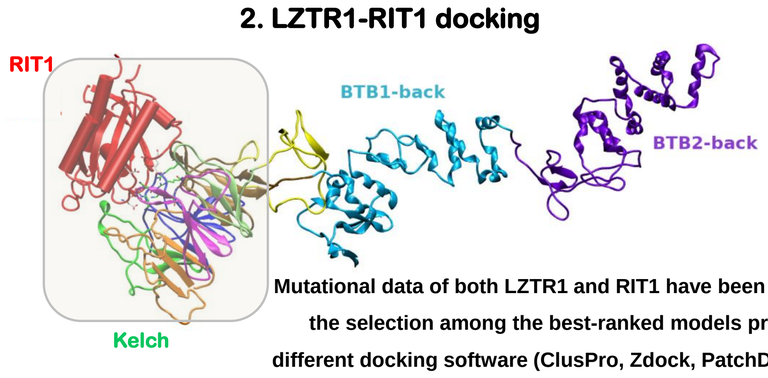


Oncogenic mutations clusterize at one face preventing LZTR1 binding

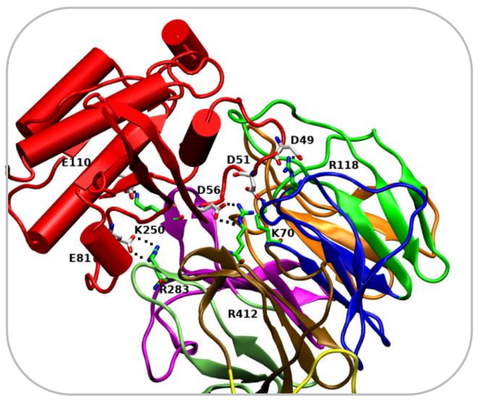


Most of pathogenic point mutations localize on the kelch repeats

2. LZTR1-RIT1 docking



Mutational data of both LZTR1 and RIT1 have been used to guide the selection among the best-ranked models predicted by different docking software (ClusPro, Zdock, PatchDock, Frodock)



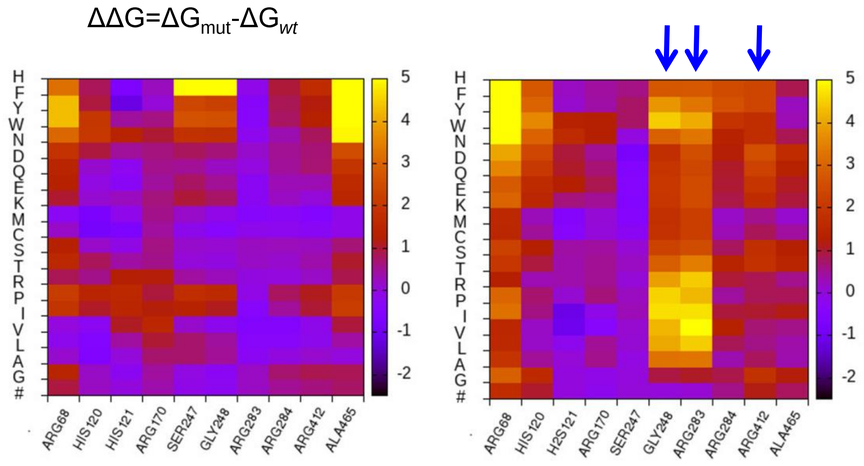
Salt bridges between **LZTR1** and **RIT1**

K70-D51
R118-D49
K250-E110
R283-E81
R412-D56

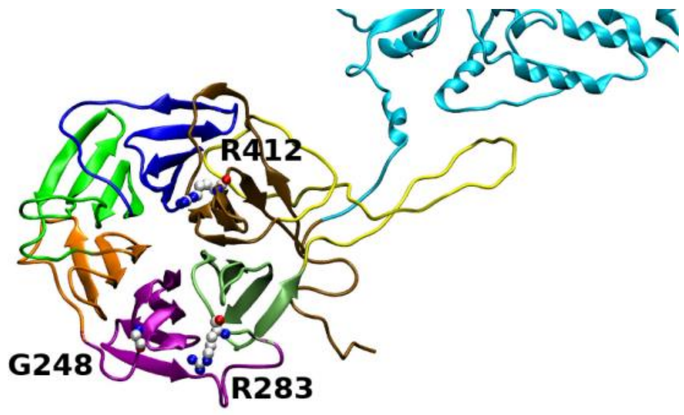
Mutational scanning for MD simulations

1. Foldx mutational analysis

$\Delta\Delta G = \Delta G_{mut} - \Delta G_{wt}$



- Position scanning is applied to free LZTR1 (left) and LZTR1 in complex with RIT1 (right). Only germline mutations of the Kelch domain are considered.
- Changes in the free energy of folding indicate that **G248, R283** and **R412** undergo large destabilization upon RIT1 binding.



R283 and R412 make ionic interactions with RIT1 in starting complex.
Pathogenicity predictions performed for 252 missense LZTR1 mutations retrieved from COSMIC and TCGA databases score **G248R**, **R283Q** and **R412C** among the most damaging variants of the Kelch domain.

2. MD simulations

run on LZTR1 in its *wild-type* and mutated sequences, in its apo form and in complex with RIT1

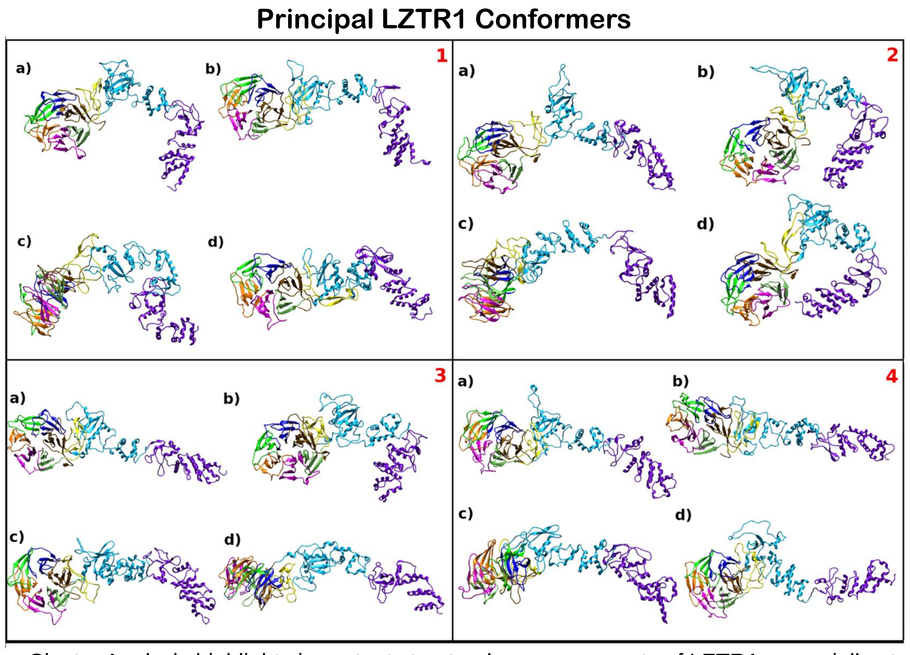
System	*wt*	G248R	R283Q	R412C
LZTR1 free	1[f]	2[f]	3[f]	4[f]
LZTR1-RIT1	1	2	3	4

MD Set Up

- *Gromacs 5.1.5*
- *Force Field: Amber03*
- *Simulation Time: 200 ns x system (x 2 replicas)*
- *8 systems (~ 270 000 atoms)*

Modulation of conformational dynamics

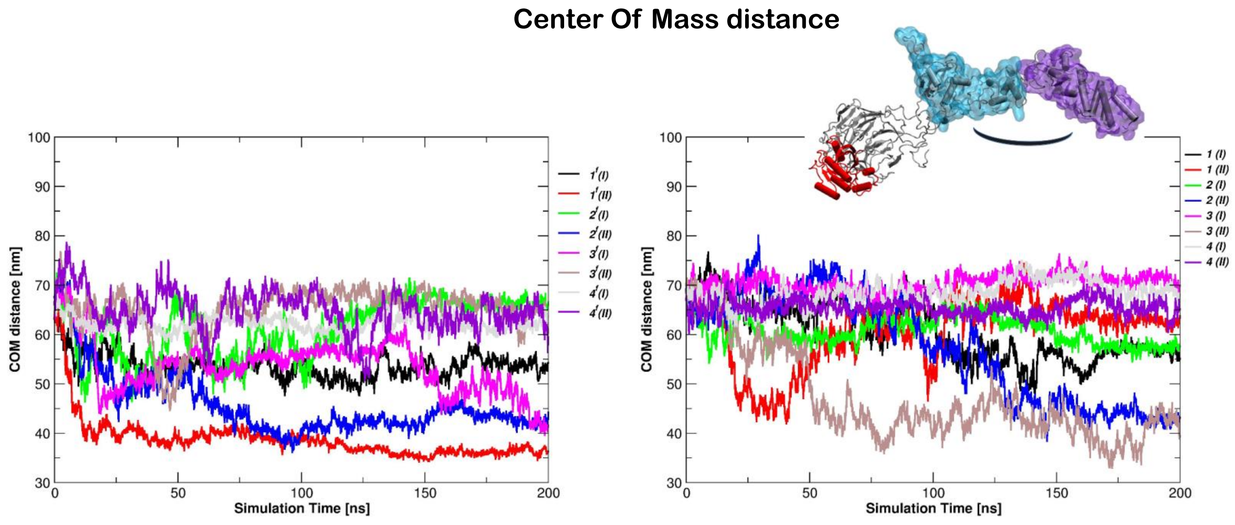
Principal LZTR1 Conformers



Structural representation of the first cluster per MD simulation:
1) *wild-type*,
2) G248R;
3) R283Q;
4) R412C.
In each panel, a-b) centroids of bound-LZTR1 in the two runs and c-d) of free-LZTR1.

- Cluster Analysis highlights important structural rearrangements of LZTR1, especially at the C-term.
- All systems, but R412C mutant (4), sample a more compact structure.
- Conformational variability is independent of RIT1 binding.

Center Of Mass distance



COM distance between the two BTB-back domains in the free LZTR1 state (left) and in LZTR1-RIT1 complexes (right).

- Most of the conformational variability converges in the **reciprocal movement** of the two BTB-back domains.
- Large fluctuation at the C-term is a common feature of LZTR1 observed for both its *wild-type* and mutated sequence and its free and bound state.
- **R412C mutant** represents the only exception: BTB-back domains do not undergo a comparable rearrangement along the simulation, either in the free or in complex with RIT1.

CONCLUDING REMARKS

The structural characterization of the interaction between the adaptor protein LZTR1 and its substrate RIT1 allows to identify key residues responsible for the recognition process. Our simulations show that LZTR1, both in the free and complexed state, experiences large structural movements and inter-domain fluctuations. While inducing a minimal repositioning of RIT1, pathogenic point mutations at the Kelch top face impact LZTR1 dynamics. In particular, R412C mutant significantly limits LZTR1 conformational variability, in either the free or bound form, suggesting a contribution not limited to the recognition mechanism. Our findings yield a mechanistic hypothesis on the role of pathogenic mutations in the recognition patterns and in the functional dynamics at the basis of E3 ubiquitin activity,[8] thus providing fundamental opportunities for drug development.

References

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